An on-chain venue trails the centralised market by half a second, in every hour measured

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Abstract

Price discovery between centralised and decentralised venues is usually studied on automated market makers. We measure it on an order-book venue: Hyperliquid, an on-chain perpetual-futures exchange whose entire trade tape is public. A grid-free Hayashi–Yoshida estimator over 758 asset-hours and four instruments puts Binance ahead of Hyperliquid by a median **575 ms** (bootstrap 95% CI $[550, 575]$), in **100% of asset-hours measured, with zero reversals**. The same estimator returns 25 ms between OKX and Binance—a factor of 23—so the lag is a property of Hyperliquid rather than of Binance.

The contribution is what the lag is *not*. Three pre-registered experiments exclude the explanation that would make it an artefact of unequal observation rates. Imposing a known lag and thinning the follower to Hyperliquid's real print times recovers that lag exactly, zero included. Independent price innovations in the follower cannot move the estimate at all—a cross-covariance is blind to them at every lag, which makes that exclusion structural rather than empirical. And a sampler coupled to price far more strongly than either venue actually is moves the peak by not one grid step. Each falsifier was written before its run, and every prediction that failed is reported as failed.

We also report a structural asymmetry between the two venues: they run on different clocks. Binance's trade arrival tracks price movement (Spearman +0.28 to +0.77 against absolute return); Hyperliquid's barely does (+0.06 to +0.17), and its quotes update roughly once per block. Whether the remaining lag is mechanical or informational is not settled here, and we state precisely what would settle it. All data, code and pre-registrations are public.

Keywords: lead–lag; price discovery; market microstructure; decentralised exchange; perpetual futures; Hayashi–Yoshida.
JEL: G14, G15, C58, C81.

1 Introduction

An on-chain perpetual-futures venue follows the centralised market by half a second. That sentence, on its own, is not a contribution: lead–lag between trading venues is among the most worked subjects in market microstructure, and finding that a newer venue trails an older one surprises nobody.

Three things make the measurement worth reporting anyway.

The first is the venue. Hyperliquid runs a central limit order book on its own chain, and its entire trade tape is public, complete and attributable—no licence, no sampling, no vendor between the analyst and the record. Most work on centralised-versus-decentralised price discovery studies automated market makers, whose pricing is mechanical rather than order-driven and whose lead–lag is therefore a different object.

The second is the universality. The lag holds in *every* asset-hour we measure—758 of them, across four instruments—with not one reversal. A lead–lag estimate that never changes sign over a month of hourly windows is not the usual shape of such a finding, and it needs an explanation of its own.

The third, and the reason this paper exists, is what the lag is not. A lag between a densely printing venue and a sparse one invites one obvious objection: that the estimator is reporting the difference in observation rates rather than a difference in information. We take that objection seriously enough to have pre-registered three experiments against it, each with its falsifier written down before the run, on a repository whose commit history dates every prediction before its result. All three exclude it. One of them excludes it *structurally*, by an argument about cross-covariance that holds whatever the data does.

We are explicit about what this paper does not claim. It does not claim the lag is exploitable: that requires knowing whether Hyperliquid's *quotes* trail or only its *prints*, which is a different measurement on different data, reported in Section 8. It does not separate mechanical latency from price discovery. And it reports, in full, one robustness check that behaved in a way we cannot explain.

2 Related work

2.1 Estimating lead–lag from asynchronous observations

Two venues trading the same instrument do not print at the same instants, and any estimator that first synchronises them onto a grid inherits the grid's resolution. That is not a minor inconvenience here: the effect under measurement is of the order of half a second, and the natural grids are of the same order or coarser.

The Hayashi–Yoshida covariance estimator [1] avoids synchronisation entirely. It sums products of returns whose observation intervals overlap, without binning, resampling or interpolation. Hoffmann, Rosenbaum and Yoshida [2] extend it to estimate the lead–lag parameter itself, by maximising the same statistic over a shift applied to one series. We use that construction throughout; Section 4 states it and reports what it was validated against.

2.2 Centralised and decentralised price discovery

3 Setting and data

3.1 Venues

Hyperliquid is a perpetual-futures exchange running a central limit order book on a purpose-built chain [3]. Order flow, fills and liquidations are recorded on-chain and published without restriction. Binance and OKX are the centralised comparison venues [4, 5], chosen because they are the two largest by perpetual volume on the instruments studied.

OKX is not decoration. A lag measured only against Binance would be consistent with Binance being unusually fast rather than Hyperliquid being slow. Measuring OKX against Binance with the identical estimator, window and code separates those two readings, and Section 5.2 reports the result.

3.2 The tape

Trade prints for all three venues come from a continuous multi-venue capture, written to hourly per-venue artefacts. Every print carries the venue's own timestamp; no clock is imposed by the recorder.

Coverage is derived *from the tape* rather than from the recorder's own status flag, and the distinction mattered. An earlier analysis dropped 49 of 193 hours on a flag that, on inspection, fired on transport reconnects rather than on recording holes. Re-deriving completeness from the data—a venue was down only if another venue kept printing through its silence—showed that **95.9% of flagged hours were in fact complete**, and that the flag is uncorrelated with market activity (Spearman +0.034 with volatility, −0.015 with event count; Mann–Whitney $p = 0.635$). The dropped hours were an unbiased subset. Recovering them widened the sample's volatility range by 38% at the top and left the headline unchanged.

3.3 Sample

Four instruments—BTC, ETH, SOL and HYPE—over a one-month window, giving 758 asset-hours after the tape-derived completeness filter, of which 191 are BTC.

The observation rates differ by roughly a factor of three, and every sparsity argument in Section 6 depends on that number: Hyperliquid's median inter-print gap on BTC is 266 ms against Binance's much shorter one. The disparity is the objection this paper has to answer, so we state it before the result rather than after.

4 The estimator, and what it was checked against

For price series observed at $\{t_i\}$ and $\{s_j\}$, the Hayashi–Yoshida statistic at shift τ is

$$U(\tau) \;=\; \sum_{i,j} \Delta X_i \, \Delta Y_j \; \mathbf{1}\{(t_{i-1}, t_i] \cap (s_{j-1} - \tau,\; s_j - \tau] \neq \emptyset\}\,, \tag{1}$$

and the lead–lag estimate is the τ maximising $|U|$. We evaluate on a 25 ms grid over ± 2 s. The sign convention is that $\tau > 0$ means Y leads X.

The estimator was validated against known answers twice, and both checks caught something.

Against imposed lags, sign included. Feeding the estimator a series shifted by a known amount recovers that amount to zero error. This check was not ceremonial: an earlier implementation returned the correct magnitude with the sign inverted, which would have been reported as *Hyperliquid leads.*

At Hyperliquid's own observation density. Validation against a known lag says nothing about behaviour at the observation rates of the actual data. Imposing a lag of zero on a series thinned to Hyperliquid's real print times returns *zero*—exactly, in every hour, on three of the four instruments, and within one grid step on the sparsest. The estimator does not manufacture a lag from unequal observation rates. Section 6 takes that further.

5 Result: a stable half-second lag

5.1 Magnitude and universality

Binance leads Hyperliquid by a median **575 ms** on BTC (bootstrap 95% CI on the median $[550, 575]$; mean 607 ms), in **100% of 191 measured hours**. Pooling the four instruments, the lag holds in **100% of 758 asset-hours, with zero reversals**.

Per instrument the medians are BTC 575 ms, ETH 550 ms, HYPE 550 ms and SOL 650 ms—a spread of 100 ms across instruments whose liquidity differs by more than an order of magnitude.

5.2 The control that makes it Hyperliquid's

Run on OKX against Binance, over the same window with the same code, the estimator returns **25 ms**—a factor of 23 smaller. The centralised venues do not lead one another to any material degree. Whatever produces the 575 ms is a property of Hyperliquid, not an artefact of using Binance as the reference.

5.3 Volatility

The lag shows no material dependence on volatility, but it is not literally invariant, and the distinction is worth making precisely because an earlier version of this work claimed the stronger form.

Pooled over 758 asset-hours and holding trade frequency constant, the partial correlation between hourly range and the estimated peak is $\rho = -0.100$ ($p = 0.006$): higher volatility, very slightly shorter lag. An effect explaining 1% of variance does not move a 575 ms median, so the result stands. The word "invariant" does not, and we withdraw it.

6 What the lag is not

Hyperliquid prints roughly three times less often than Binance. The objection this invites is that the estimator reports that disparity rather than any difference in information. This section answers it with three experiments, each pre-registered with its falsifier written before the run.

6.1 Not an artefact of unequal observation rates

Take Binance's own series, shift it by a *known* lag L, and then thin it to Hyperliquid's observed print times for the same hour—producing a follower exactly as sparse as Hyperliquid, with a lag we chose. The unmodified estimator must return L.

It does, for $L \in \{0, 100, 300, 575, 1000\}$ ms, exactly and in every hour on BTC, ETH and HYPE, and within one 25 ms grid step on SOL, the sparsest instrument.

The line that decides it is $L = 0$. Imposing no lag and thinning to Hyperliquid's cadence returns zero—not a drift toward 575 ms, not a drift toward anything. Whatever produces the measured lag, it is not the estimator inventing one from unequal observation rates.

6.2 Not the follower's own price innovations

Thinning a deterministic copy of the leader is a weaker test than it looks: the follower there has no price of its own. Hyperliquid does. We therefore repeat the construction with a follower that carries an independent price component of calibrated size, and vary that size.

The estimate does not move—at noise ratios up to eighty times the leader's own variance.

This exclusion is *structural* rather than empirical, and that is the point. The statistic is a cross-covariance, so if the follower is $X(t - L) + W$ with W independent of X, then $\mathbb{E}[\Delta X \cdot \Delta W] = 0$ at every lag and W contributes nothing to $U(\tau)$ at any τ. Independent noise moves the estimator's variance, never its peak. The measurement confirms an algebraic fact rather than establishing an empirical one.

6.3 Not endogenous sampling

Both constructions above share a defect: their observation times are independent of the follower's own price movements, by construction. Real venues are not sampled that way—a venue prints *because* a trade happened, and trades happen when the price is moving.

We therefore sample the synthetic follower on a total-variation clock: an observation is emitted whenever cumulative absolute log-price movement on the follower's own path crosses a threshold, with the threshold set so the count matches Hyperliquid's real count for that hour. Same sparsity, different reason for it.

The peak does not move—not one 25 ms grid step, at any instrument or dimension. And the test is harsher than reality: the total-variation clock samples on nothing *but* the series' own movement, where Hyperliquid's measured coupling to price is between +0.06 and +0.17 (Section 7). The null is an upper bound.

6.4 What that leaves

Observation density is excluded on three independent constructions, one of them structurally. What remains as candidate explanations is block cadence, network latency, and genuine price discovery. Across instruments the correlation between trade frequency and lag is $\rho = -0.656$, which has the signature of a density effect; with no estimator-side mechanism surviving, we attribute it to market structure—sparser instruments are genuinely slower—rather than to the measurement.

6.5 One check we cannot explain

Honesty requires reporting the following. Thinning the *leader* to Hyperliquid's cadence moves the measured peak upward on the real pair, by between 25 and 212 ms depending on instrument. No synthetic construction we built reproduces that behaviour.

That manipulation degrades the leader, and the reported measurement never does—Binance enters at full density throughout. So this is an unexplained property of a deliberately degraded measurement, not of the result. We state it because it is unexplained, and we say what would reopen it: a quote-based measurement returning a materially different magnitude.

7 Two venues, two clocks

The endogenous-sampling test above required measuring how strongly each venue's printing couples to price movement. That measurement is a finding in its own right.

For each one-second bin we correlate a venue's print count with the absolute log return over the same second, using Binance's return as the movement proxy for both venues so that no venue's activity is measured against its own observed returns:

	Hyperliquid	Binance
HYPE	+0.167	+0.771
BTC	+0.126	+0.447
ETH	+0.112	+0.618
SOL	+0.057	+0.280

Binance prints markedly more when the market moves. Hyperliquid barely does—its coupling is three to five times weaker on every instrument.

A likely reason is measurable and we report it as consistent rather than as tested: Hyperliquid's quotes update about 7.9 times a second on BTC, roughly once per block. Trade arrival on an on-chain venue is gated by block production; on a centralised venue it is gated by order flow. The two venues are sampled by different clocks, and that is a structural difference between on-chain and off-chain market structure rather than a property of this window.

8 Quotes or prints?

Everything above is measured on trade prints. Trades and quotes answer different questions, and the difference decides what the result means.

If Hyperliquid's *quotes* trail Binance's by half a second, then resting orders sit at stale prices and the lag is something an informed participant could act on, subject to fees. If only its *prints* trail, then the last traded price is merely old—the book may be perfectly current—and there is nothing there to take.

Three facts about that measurement are already established and hold whatever it returns.

Recorded depth books cannot answer it. Binance publishes partial order-book depth on a venue-imposed 500 ms grid. A lag of the order of 575 ms cannot be resolved against a series published on a 500 ms clock, whatever the recording cadence downstream.

The best-bid-offer channels can. Both venues publish a BBO stream that pushes on every change with no interval. A capture of those channels began on 2026-07-28.

It is the harder measurement, not the easier one. On quotes the two venues differ in observation rate by a factor of 15 to 40, against roughly 3 on trades. The estimator recovers known lags exactly at those densities, including zero, so the instrument is validated for the regime; but the asymmetry the result must survive is an order of magnitude worse than the one addressed in Section 6.

9 Implications

For empirical work, the practical consequence is narrow and concrete: an on-chain trade price is not a contemporaneous observation of the market price. Any study using such a series alongside centralised data—realised volatility, correlation, event studies around announcements—inherits a half-second offset that is stable enough to correct for and large enough to matter at intraday horizons.

For market structure, the two-clocks result is the more interesting one. It says the difference between these venues is not only speed but the *sampling mechanism*: one venue's tape is driven by order flow, the other's by block production. That distinction survives whatever the quote measurement returns.

We deliberately do not draw an execution implication. That requires Section 8 to show stale quotes *and* the move to exceed costs; Hyperliquid's taker fee at the base tier is 4.5 basis points, which is the number any such claim would have to clear.

10 Limitations

The unexplained check. Leader-thinning inflates the measured peak on the real pair and no synthetic construction reproduces it. Stated in Section 6; unresolved.

One estimator. Hayashi–Yoshida is validated here against known answers, at the data's own observation density, and under three adversarial samplings. It remains one estimator, and a result confirmed by a second family would be stronger.

Scope. Four instruments, one venue pair plus one control, one month. The universality claim is over that sample and no wider.

Trade prints only, until Section 8 is complete.

No submission-level instrument. Nothing in this data sees an order being submitted, only executed. "Price discovery" is therefore inferred from execution timing, and the mechanical component—block cadence, propagation—is not separated from the informational one.

11 Reproducibility

Every experiment behind this paper is pre-registered in a public repository, in a file dated and committed before the corresponding run. Predictions that failed are reported as failed, and several did: an estimator sign inversion, a claim of volatility invariance now withdrawn, a sample-selection concern that measured to zero, and a mechanism attribution retracted when its timing signature did not appear. The data files, the decoders and the analysis code are public.

Declarations

Funding. None.

Competing interests. The trade tape analysed here is recorded by market-data infrastructure the author operates. No trading position was held in any instrument studied during the window.

Data availability. All data, code and pre-registrations are public.

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